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Data Analytics with Tableau Assignment

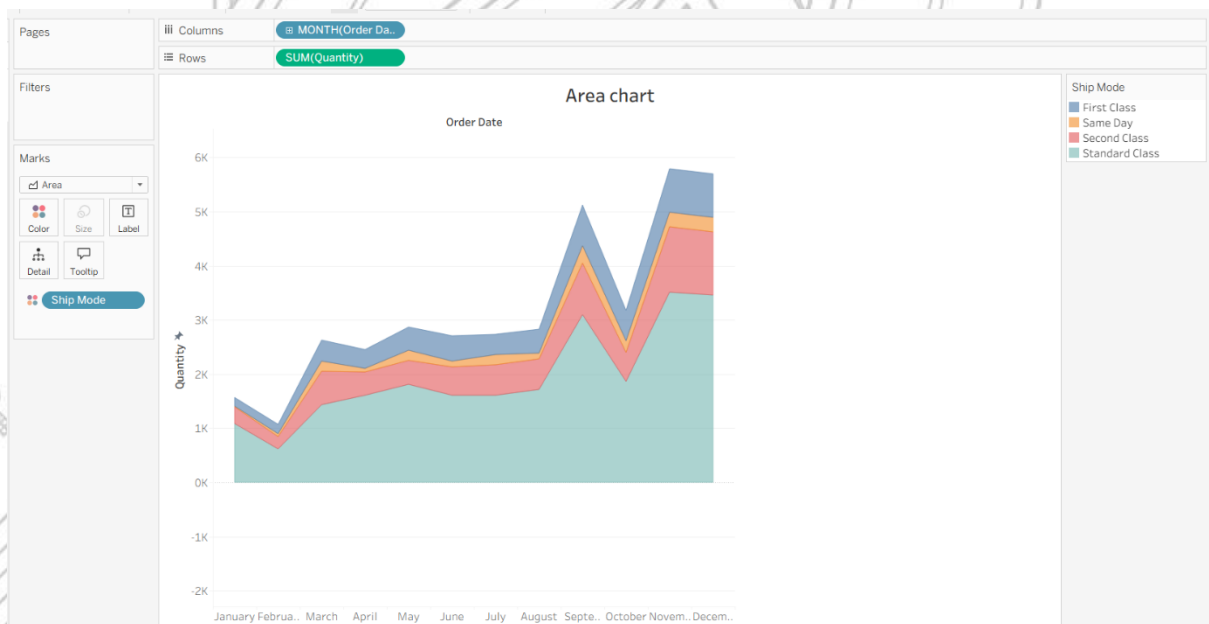
QUESTION: Prepare the following charts

Sheet - 01 – Area chart

Steps:

1. Drag **Order Date** to the **Columns** shelf and change the date level to **Month** by right-clicking it.
2. Drag **Quantity** to the **Rows** shelf.
3. Place **Ship Mode** on the **Color** option in the **Marks** card.
4. In the **Marks** card, open the mark type drop-down and select **Area** to create the area chart.

Output



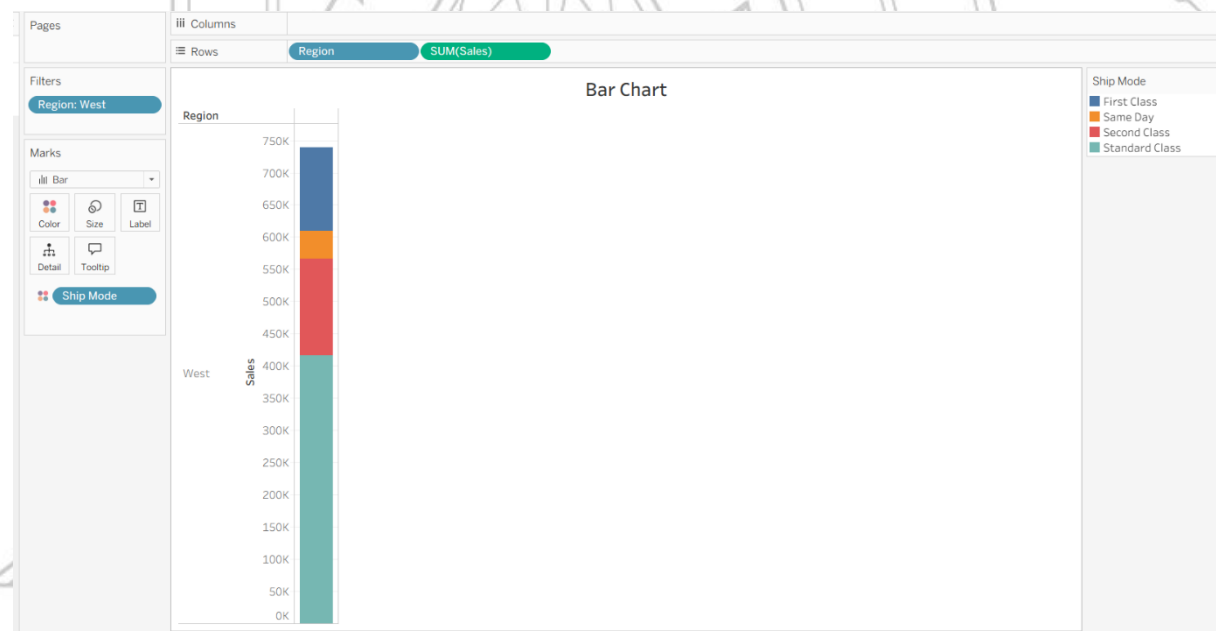
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Sheet – 02 – Bar Chart

Steps:

1. Drag **Sales** to the **Rows** shelf.
2. From the **Marks** card drop-down menu, select **Bar** as the chart type.
3. Drag **Ship Mode** to the **Color** option in the **Marks** card.
4. Drag **Region** to the **Rows** shelf and place it before **Sales**.
5. Drag **Region** to the **Filters** shelf and choose **West** to display only West region data.

Output:



Sheet – 03 – Box plot

Steps:

1. Place **Sales** on the **Rows** shelf to display sales values.
2. Change the **Marks** type to **Bar** from the Marks card.
3. Drag **Ship Mode** to **Color** to distinguish each shipping method.
4. Add **Region** to the **Rows** shelf and position it before **Sales**.
5. Use **Region** as a filter and select **West** to view data for the West region only.

Output:

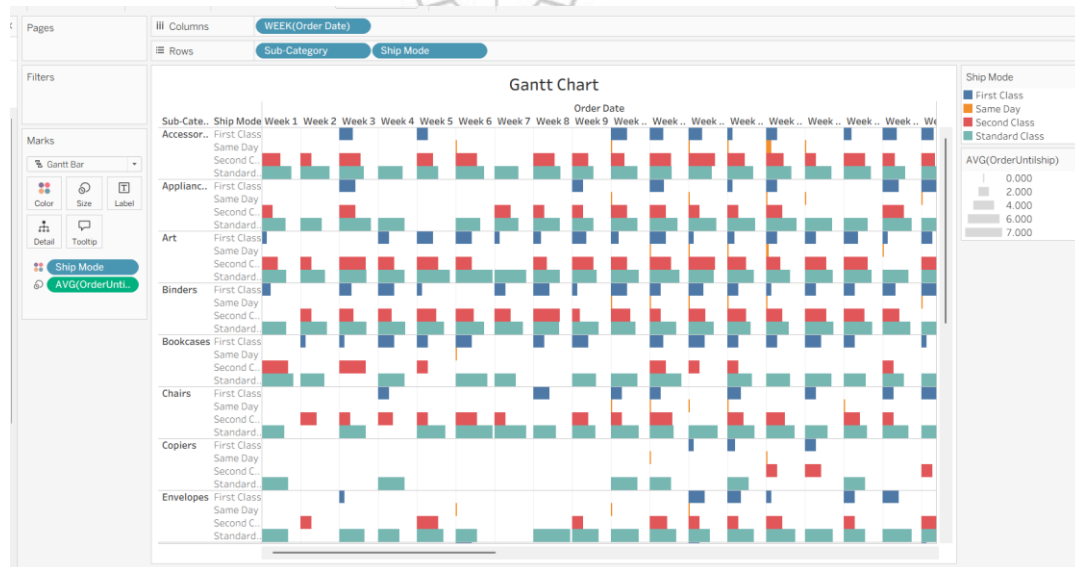


Sheet – 04 – Gantt chart

Steps:

1. Drag **Order Date** to the **Columns** shelf.
2. Open the **Order Date** drop-down menu and select **More → Week Number** to display data by week.
3. Drag **Sub-Category** to the **Rows** shelf, then place **Ship Mode** next to it.
4. Change the **Marks** type to **Gantt Bar** from the Marks card.
5. Create a calculated field named **OrderUntilShip** using the formula:
DATEDIFF('day', [Order Date], [Ship Date])
6. Drag **OrderUntilShip** to **Size** on the Marks card.
7. Right-click **OrderUntilShip** on the Marks card and change its aggregation to **Average**.
8. Drag **Ship Mode** to **Color** on the Marks card to differentiate the Gantt bars by shipping method.

Output:

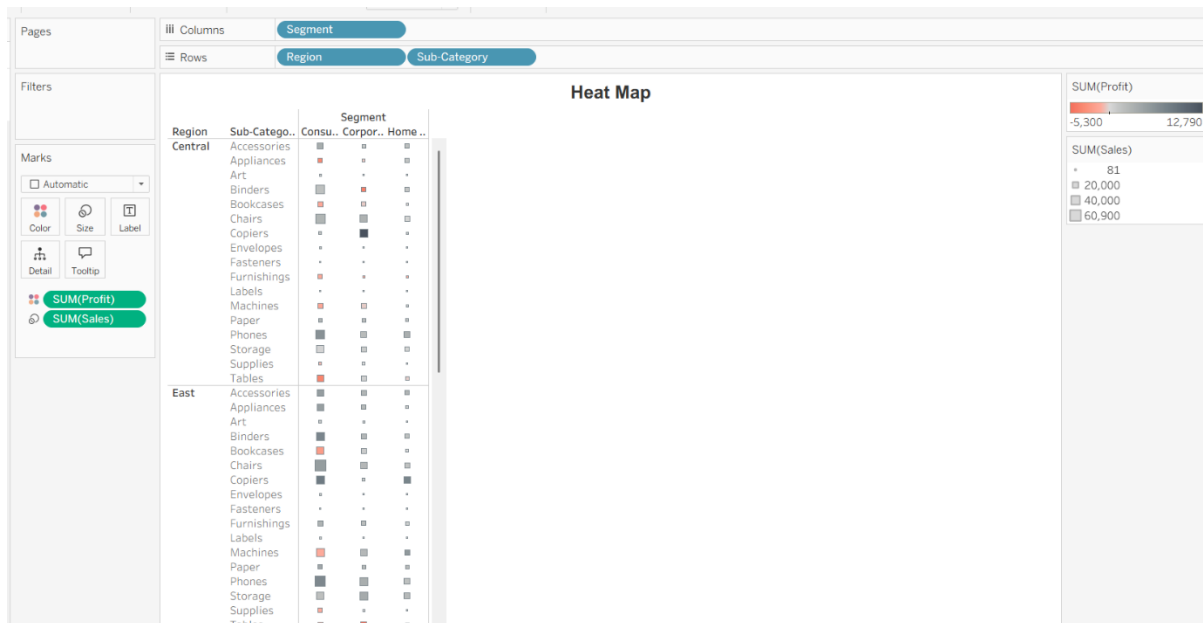


Sheet – 05 – Heat Map

Steps:

1. Drag **Segment** to the **Columns** shelf and place **Region** and **Sub-Category** on the **Rows** shelf.
2. Drag **Profit** to **Color** on the **Marks** card and choose the **Red-Black Diverging** color palette.
3. Drag **Sales** to **Size** on the **Marks** card and adjust the size slider to fit the boxes within the cells.
4. Click **Color** on the **Marks** card and set the **Border** color to **Medium Gray** for better visual clarity.

Output:



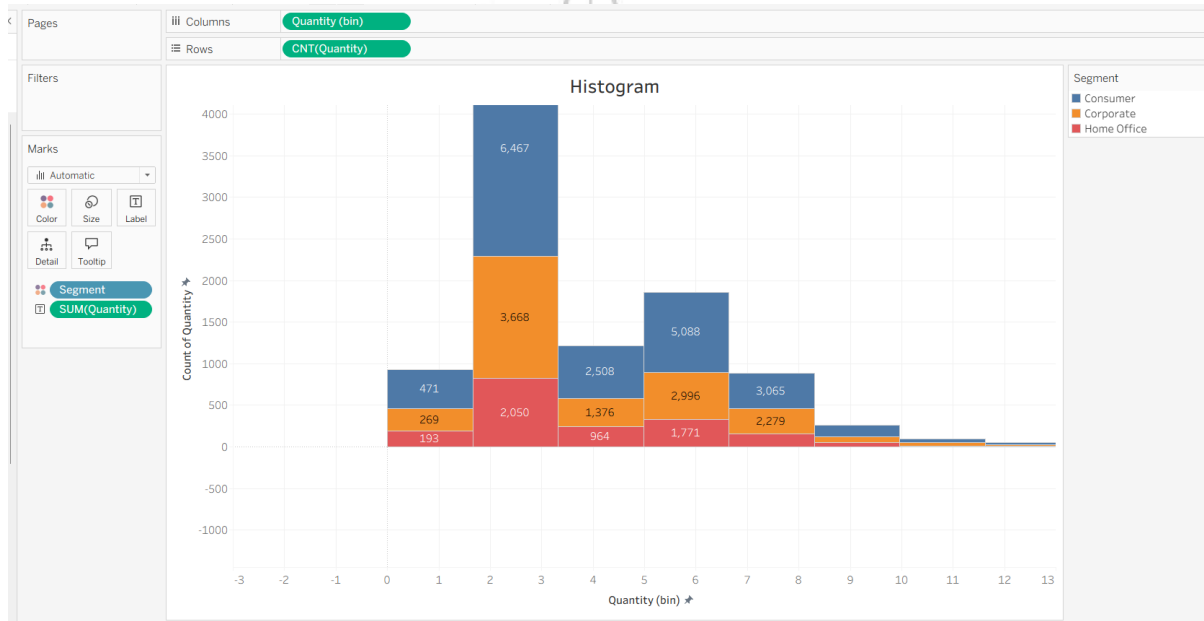
Sheet – 06 – Histogram

Steps:

1. Drag **Quantity** to the **Columns** shelf and select **Histogram** from the **Show Me** panel to automatically generate bins and frequency counts.
2. Drag **Segment** to **Color** on the **Marks** card to create a stacked histogram by segment.
3. Hold **Ctrl** and drag **CNT(Quantity)** from the **Rows** shelf to the **Label** option on the Marks card.
4. Right-click **CNT(Quantity)** on the **Marks** card, choose **Quick Table Calculation**, and select **Percent of Total** to display percentages.

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Output:



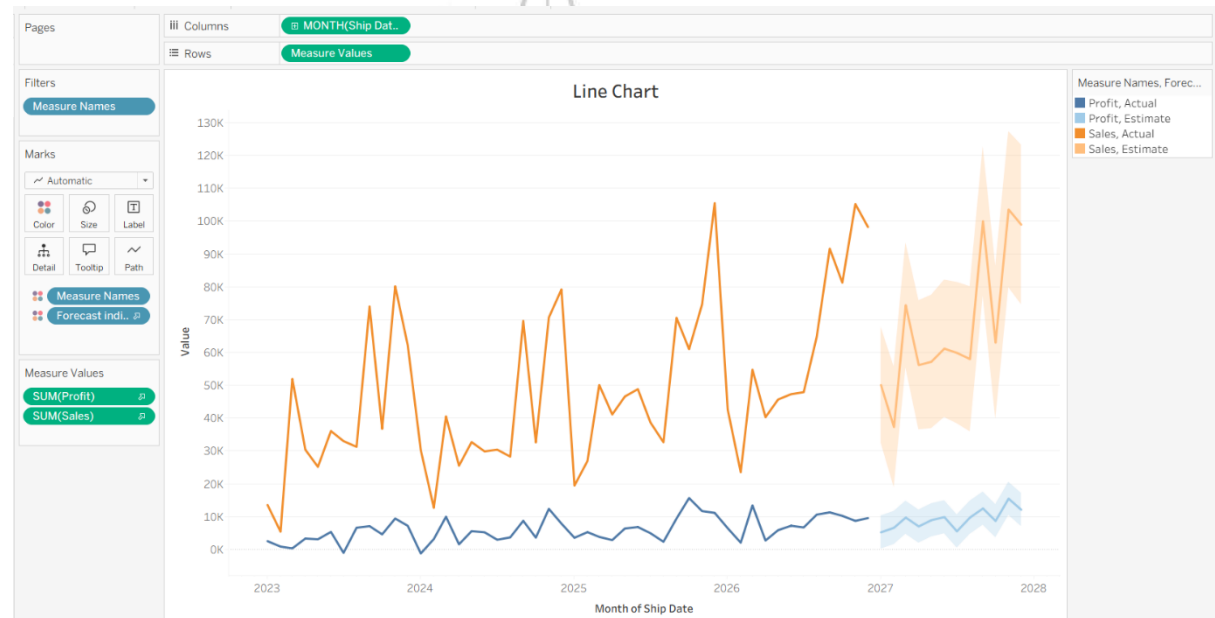
Sheet – 07 – Line Chart

Steps:

1. Drag **Ship Date** to the **Columns** shelf and change it to **Month (Continuous)** from the lower section of the date drop-down menu.
2. Drag **Sales** to the **Rows** shelf, then drag **Profit** directly onto the existing **Sales** axis to create a blended axis chart.
3. Ensure **Measure Names** is placed on the **Color** option in the **Marks** card to differentiate the **Sales** and **Profit** lines.
4. Open the **Analytics** pane, drag **Forecast** into the view, and drop it onto the chart to display predicted future trends.

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Output:

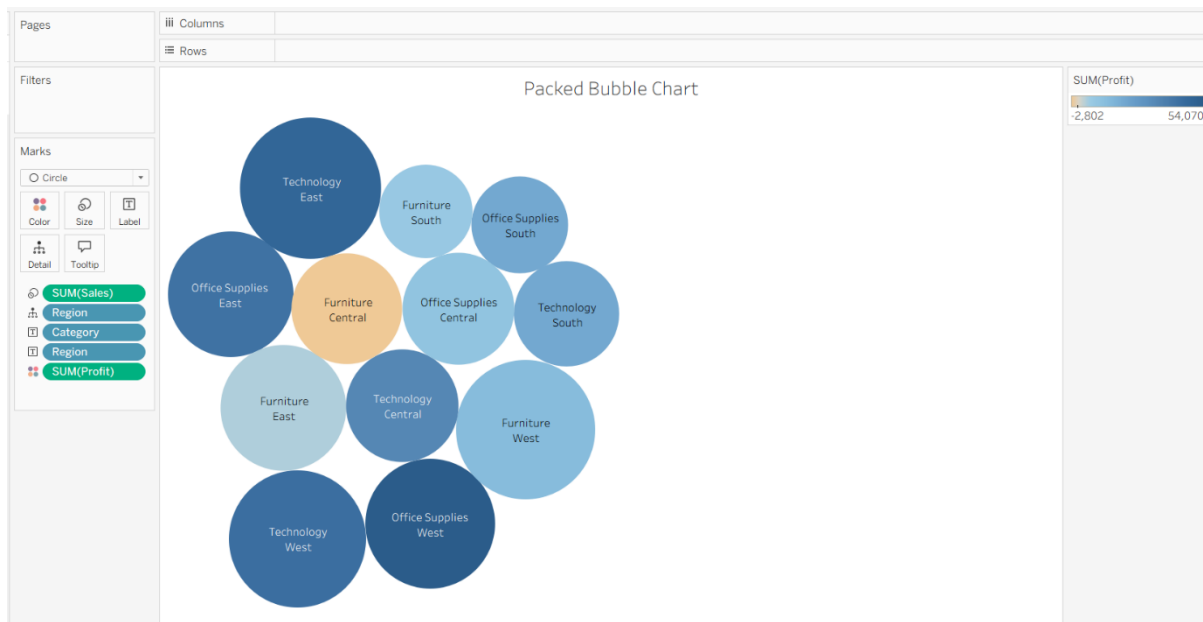


Sheet – 08 – Packed Bubble Chart

Steps:

1. Drag **Category** to the **Columns** shelf and **Region** to the **Rows** shelf, then select **Packed Bubbles** from the **Show Me** panel.
2. Drag **Sales** to **Size** on the **Marks** card to control the size of each bubble.
3. Drag **Profit** to **Color** on the **Marks** card to represent profitability using color variations.
4. Drag both **Category** and **Region** to **Label** on the **Marks** card so each bubble displays its corresponding data segment.

Output:

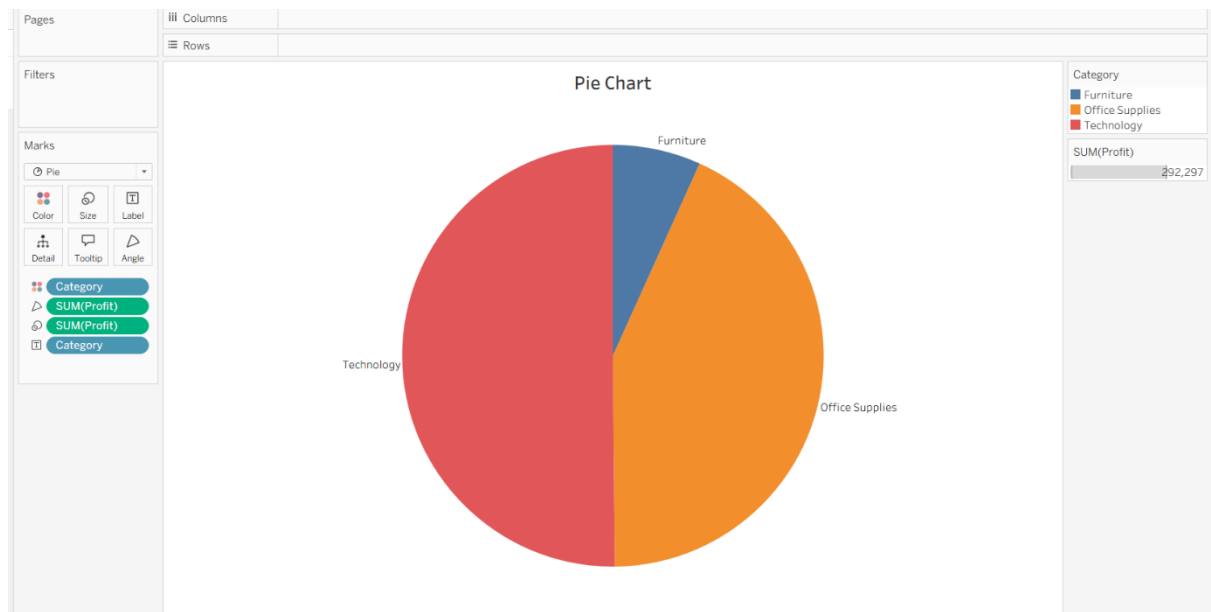


Sheet – 09 – Pie Chart

Steps:

1. Drag **Profit** to the **Columns** shelf and **Category** to the **Rows** shelf, then choose **Pie Chart** from the **Show Me** panel.
2. Place **Category** on the **Color** option in the **Marks** card to create separate slices for each category.
3. Drag **Profit** to the **Angle** mark to determine the size of each pie slice.
4. Drag **Category** to the **Label** mark so that each slice displays its category name.

Output:



Sheet – 10 – Scatter Plot

Steps:

1. Drag **Sales** to the **Columns** shelf and **Profit** to the **Rows** shelf.
2. Drag **Category** to **Color** and **Region** to **Detail** on the **Marks** card to create multiple data points.
3. Open the **Analytics** pane, drag **Trend Line** into the view, and choose the **Linear** model.
4. Click **Shape** on the **Marks** card, select **Open Circle** shapes, and adjust the transparency to match the desired output style.

Output:



Sheet – 11 – Text Table with Percentages

Steps:

1. Drag **Order Date** to the **Columns** shelf and place **Region** and **Sub-Category** on the **Rows** shelf.
2. Drag **Sales** to the **Text (ABC)** mark on the **Marks** card to populate the table with sales values.
3. Right-click **SUM(Sales)** on the **Marks** card and select **Quick Table Calculation** → **Percent of Total**.
4. Right-click **SUM(Sales)** again, choose **Compute Using**, and select **Pane (Down)** to calculate percentages within each region.

Output:

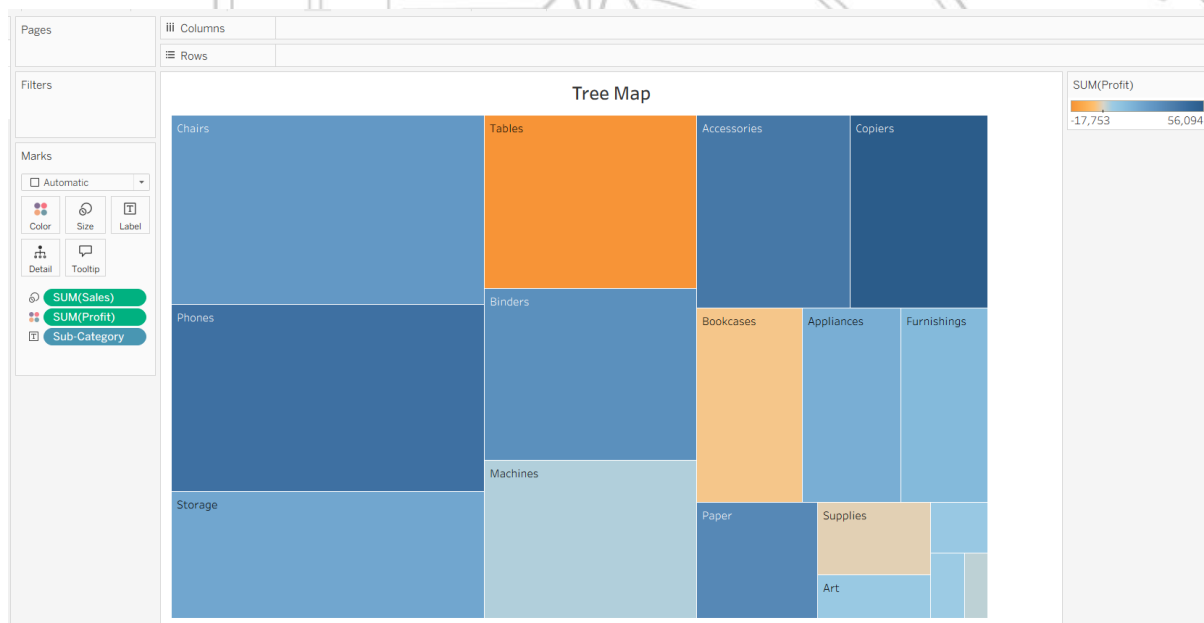
Text Table With Percentage

		Order Date			
		2023	2024	2025	2026
Region	Sub-Category				
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
East	Storage	10.68%	8.10%	8.63%	9.28%
	Supplies	0.42%	0.32%	2.89%	2.98%
	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%
	Bookcases	8.02%	12.51%	3.30%	3.42%
	Chairs	16.89%	12.75%	12.70%	14.50%
	Copiers	3.91%	3.16%	10.81%	11.27%
	Envelopes	0.99%	0.56%	0.71%	0.39%
	Fasteners	0.16%	0.14%	0.21%	0.13%
	Furnishings	3.32%	3.46%	5.43%	5.07%
	Labels	0.38%	0.33%	0.47%	0.42%
	Machines	10.81%	9.18%	13.80%	5.46%
	Paper	3.25%	2.53%	2.71%	3.62%
	Phones	16.04%	14.50%	11.76%	16.79%
	Storage	12.07%	8.48%	12.11%	9.40%

Sheet – 12 – Treemap

Steps:

1. Drag **Sub-Category** to the **Columns** shelf and **Sales** to the **Rows** shelf, then select **Treemap** from the **Show Me** panel.
2. Place **Sub-Category** on the **Label** option in the **Marks** card to identify each rectangle.
3. Drag **Profit** to **Color** and **Sales** to **Size** on the **Marks** card to control the color and size of the treemap rectangles.
4. **Output:**

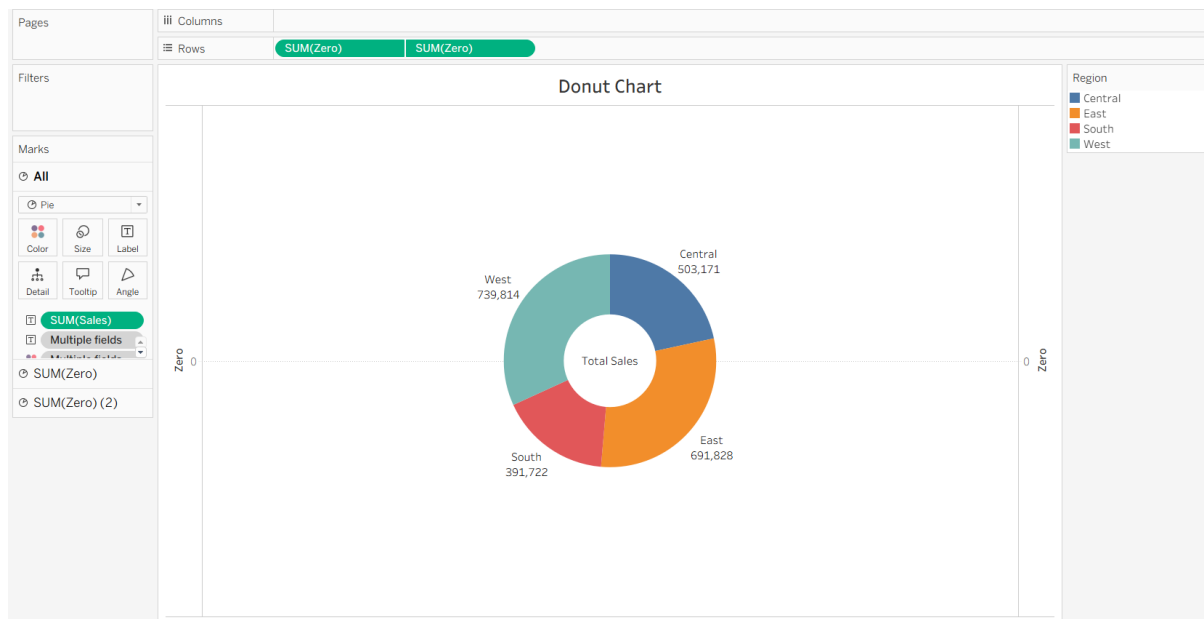


Sheet – 13 – Donut chart

Steps:

1. Create a calculated field named **Zero** with the formula 0, drag it twice to the **Rows** shelf, and set the **Marks** type to **Pie**.
2. On the first **SUM(Zero)** Marks card, drag **Region** to **Color** and **Sales** to **Angle**. On the second **SUM(Zero)** Marks card, remove all fields and set the color to **White**.
3. Right-click the second **SUM(Zero)** pill on the **Rows** shelf and select **Dual Axis**, then increase the size of the first pie chart to form the donut ring.
4. Select the second **Marks** card and drag **Sales** to **Label** to display the total value in the center of the donut chart.

Output:

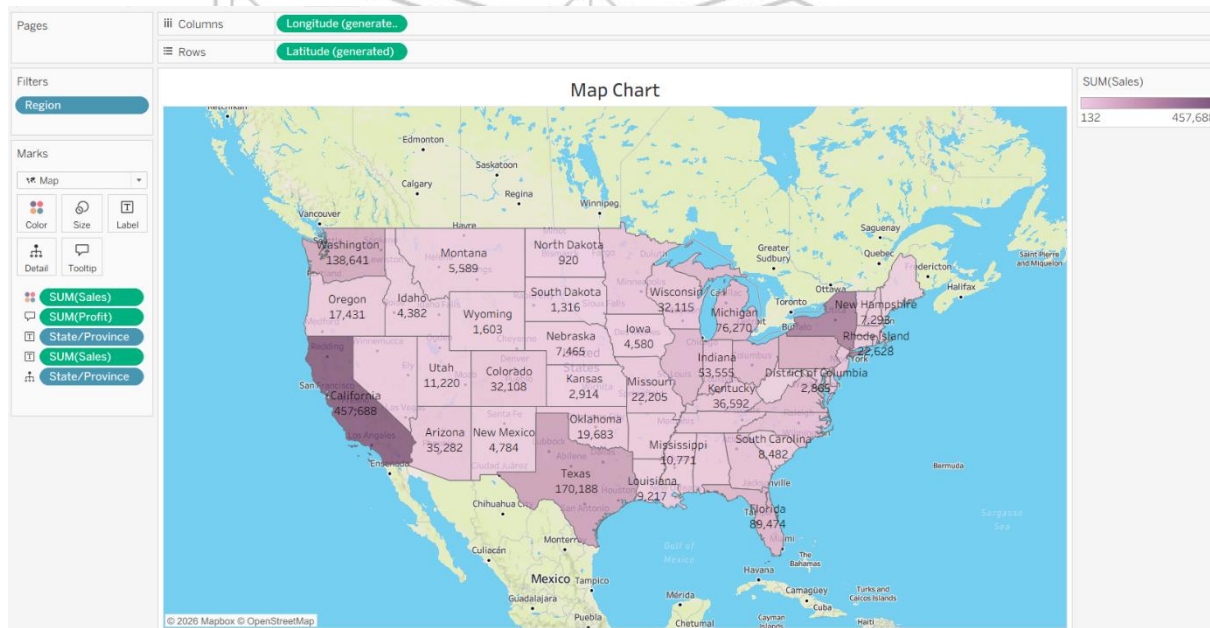


Sheet – 14 – Map Chart

Steps:

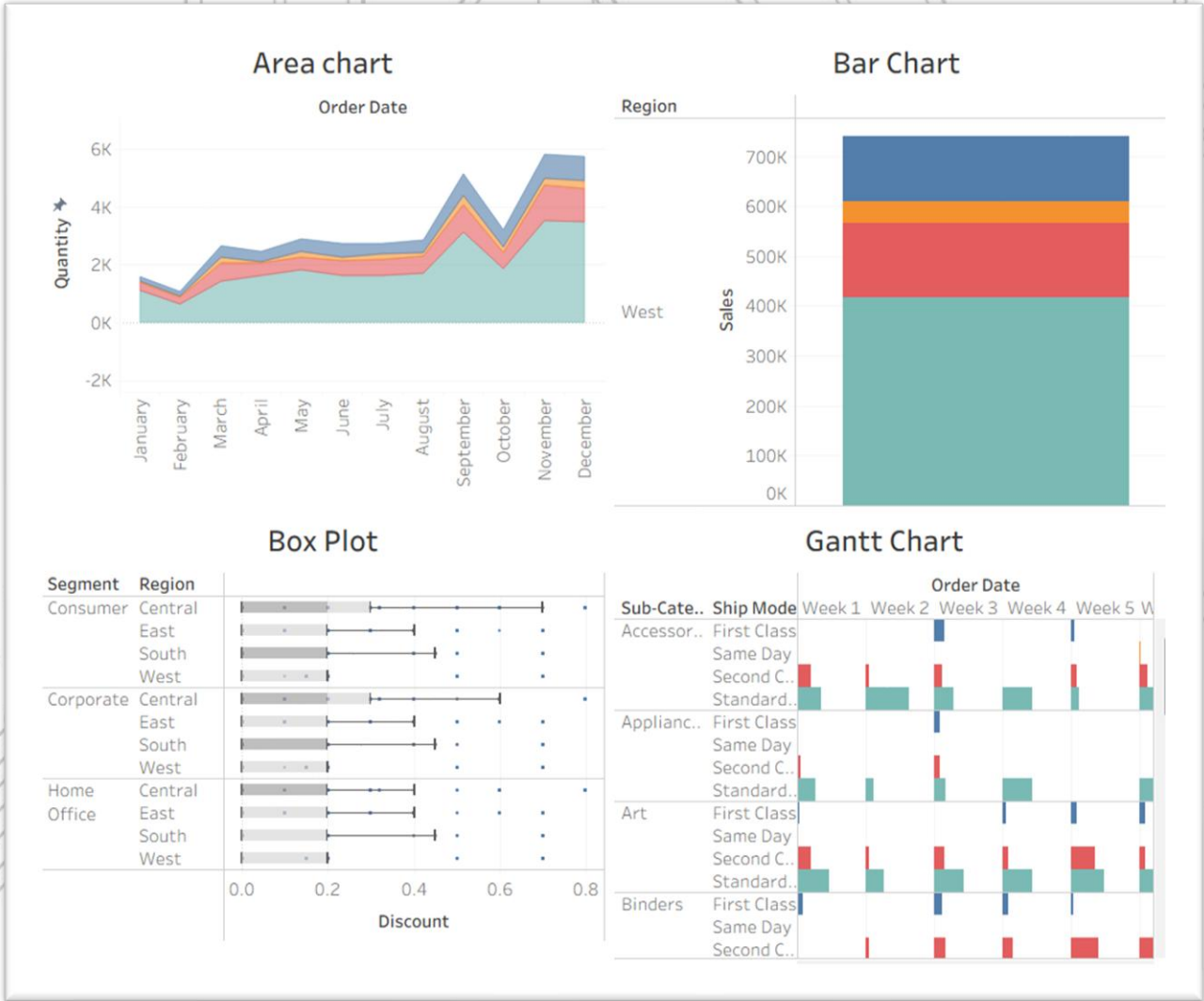
1. Double-click **State** (or drag it to the view) to automatically create a map using generated **Latitude** and **Longitude**.
2. Change the **Marks** type to **Map (Filled Map)** and drag **Sales** to **Color**, then apply a **Purple** color palette.
3. Drag **State** and **Sales** to **Label**, and place **Region** on the **Filters** shelf for interactive filtering.
4. Go to **Map → Map Layers**, select **Outdoors** as the background style, and set **Washout** to **70%** for improved readability.

Output:

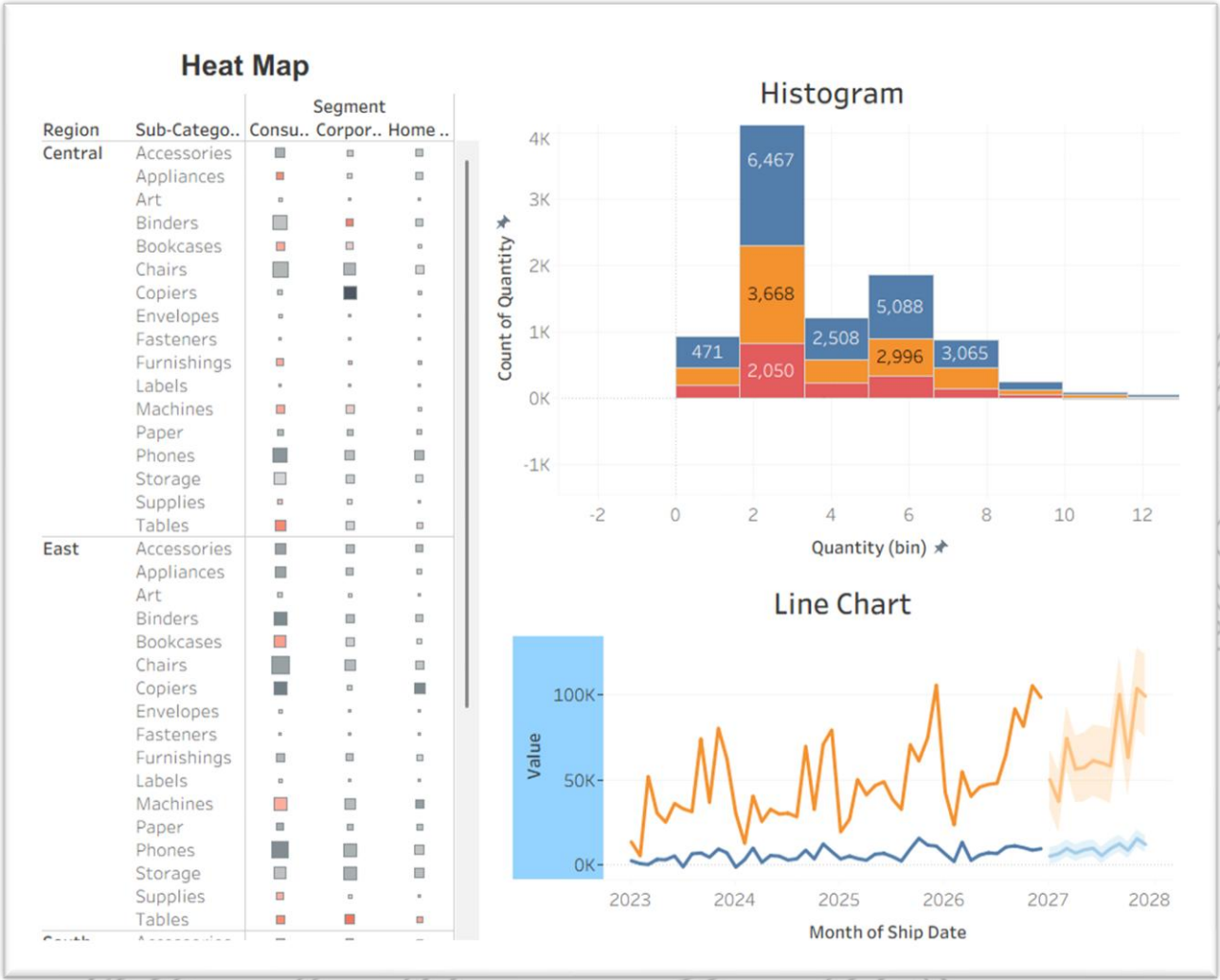


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DASHBOARDS:
Dashboard – 01

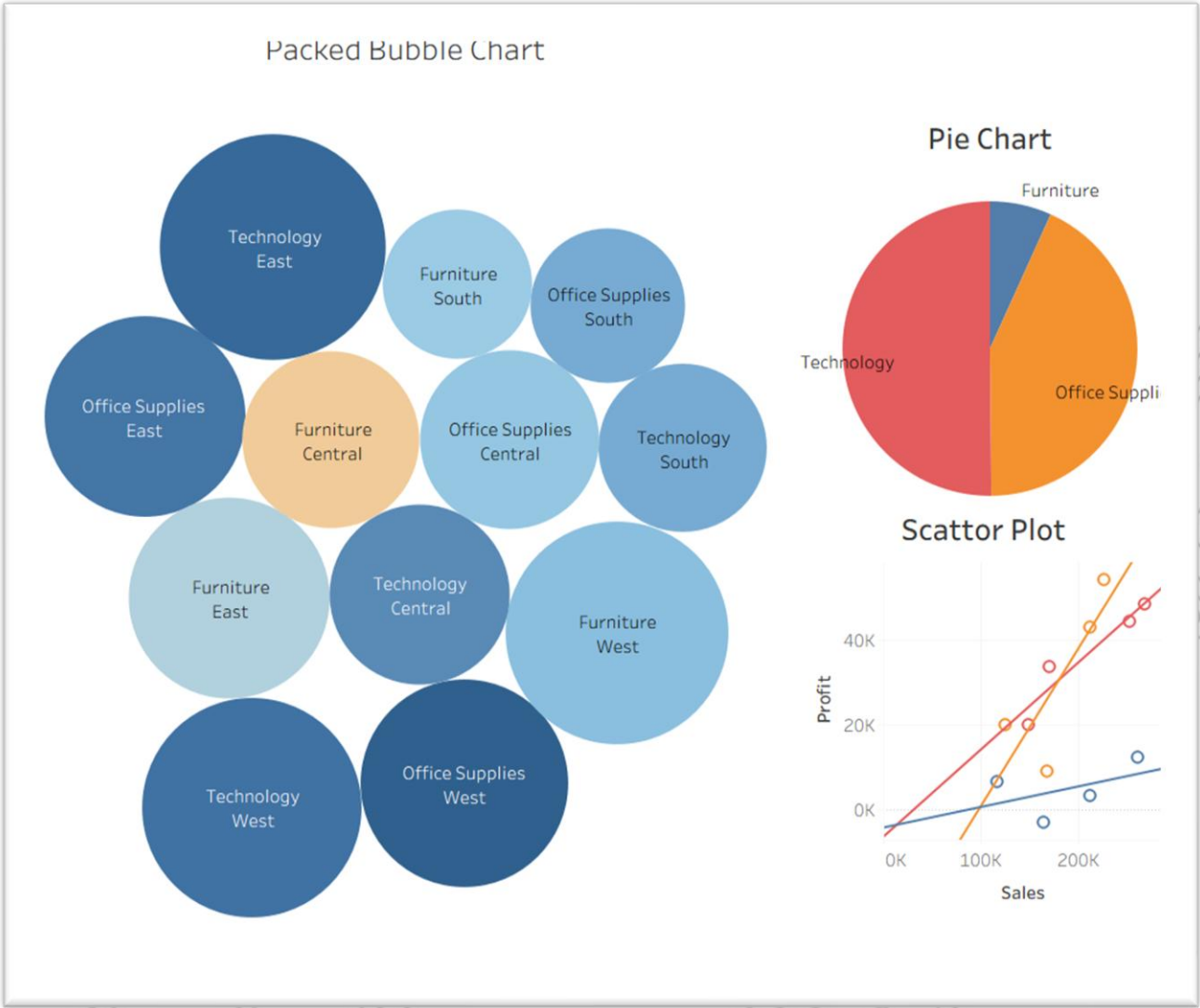


Dashboard – 02



DELTA
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Dashboard – 03



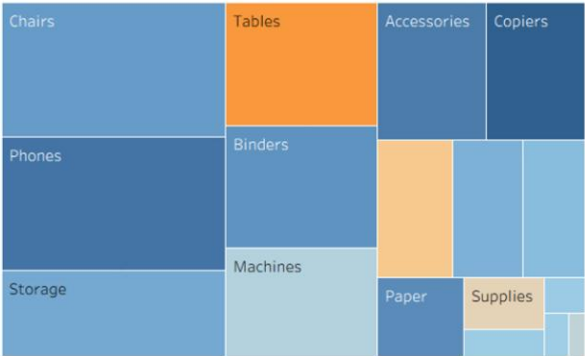
DELTA
- ACHIEVERS -

Dashboard – 04

Text Table With Percentage

Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
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	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
	Storage	10.68%	8.10%	8.63%	9.28%

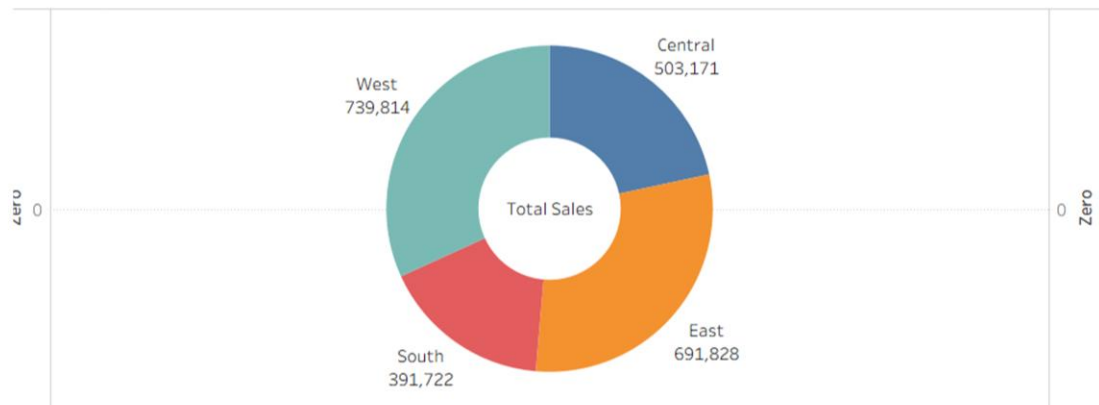
Tree Map



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Dashboard – 05

Donut Chart



Map Chart



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